

What is MIME?

- **MIME**, also known as the **Multipurpose Internet Mail Extensions**, is an Internet standard introduced in 1991.
- It allows us to **send, receive and view e-mail messages that supports audio, images, video, applications and texts written in character sets other than ASCII.**

What is S/MIME?

- S/MIME means **Secure/Multipurpose Internet Mail Extensions**.
- It is a **technology that allows us to encrypt the content of our e-mails, so that they are not vulnerable to cyber-attacks.**
- S/MIME was first developed by the **RSA Data Security** to ensure the security of e-mail messages, then it became a standard with the help of IETF.
- S/MIME is based on **asymmetric encryption and public key infrastructure**.
- It aims to provide a layer of security for the e-mail messages with the help of **encryption and authentication techniques**.

- In other words, S/MIME makes it possible for you to **sign your e-mails digitally** so that only the intended receiver of your e-mails can receive and view them. Also, S/MIME makes sure that nobody alternates the content of your e-mail while it is on its way to the receiver's inbox.

Working of S/MIME

1. Authentication in S/MIME

- Authentication is an essential security service that must be provided in S/MIME to ensure that the E-mail has come from a legitimate user.

Sender side:

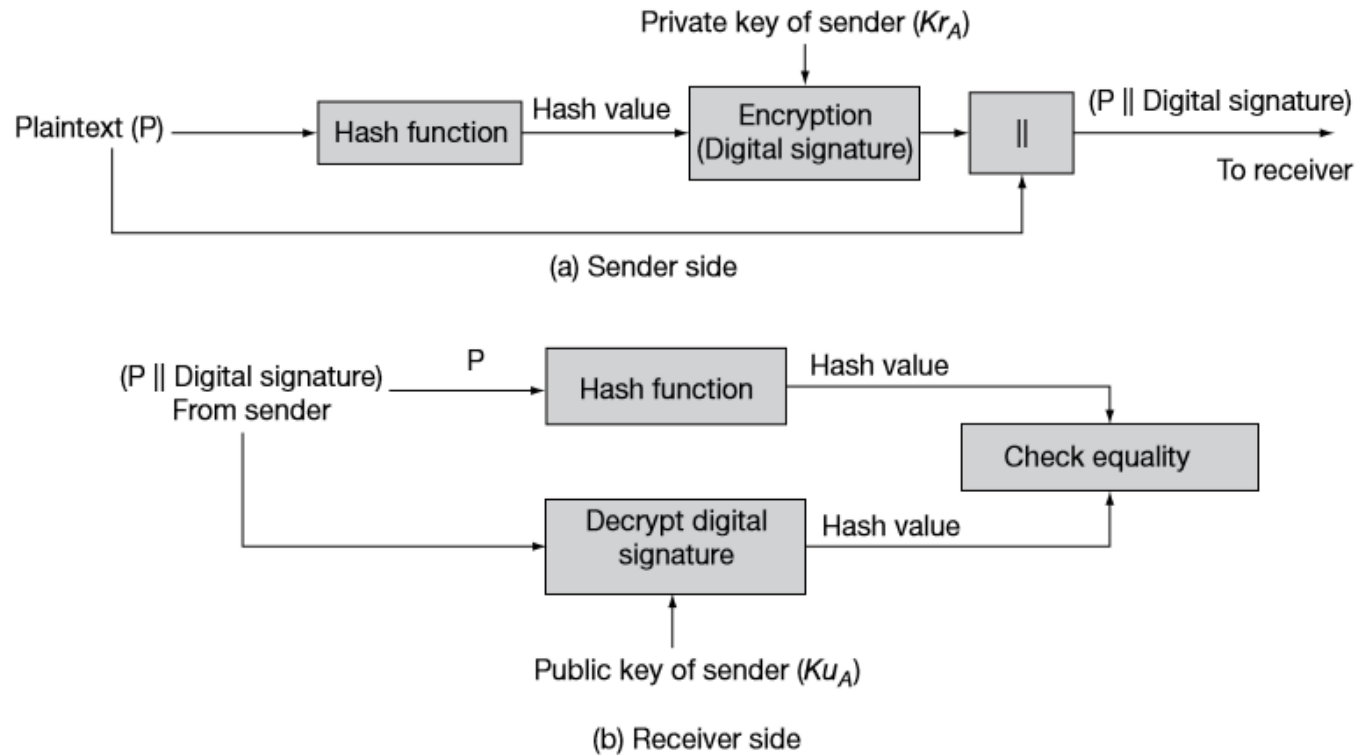
1. The sender's message is denoted as **plaintext (P)** and the sender hashes the plaintext message (E-mail) using the hash function to produce the hash value.
2. This hash value is encrypted by the sender using its **private key (K_{r_A})** to produce the digital signature.
3. The plaintext of the sender together with the digital signature is then sent to the receiver.

Receiver side:

1. The receiver uses the same hash function on the plaintext to produce the same hash value.

2. The receiver uses the **public key of the sender (Ku_A)** to decrypt the digital signature to produce the hash value.
3. Finally, the receiver checks the equality of both the newly computed hash value and already received hash value to authenticate the plaintext.

The figure below illustrates the authentication service at sender and receiver side.



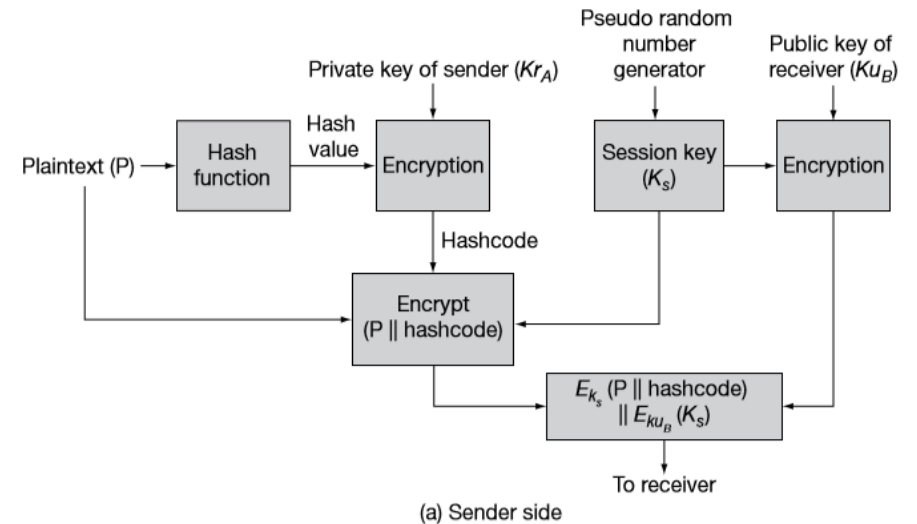
2. Authentication and Confidentiality in S/MIME

Confidentiality is also an essential security service that must be ensured when an E-mail is sent from one user to another user.

Sender side:

1. The sender wants to communicate an E-mail (plaintext) with the receiver securely. The sender hashes her message using the hash function to produce the hash value.
2. The sender uses its private key (Kr_A) to encrypt the hash value to produce the hash code.
3. The sender chooses a 128-bit session key (K_s) to encrypt the plaintext of the sender together with the hash code.
4. The sender encrypts the session key using the public key of the receiver (Ku_B) and produce the ciphertext as

$$E_{K_s}(P \parallel \text{hashcode}) \parallel E_{Ku_B}(K_s).$$



Receiver side:

1. After receiving the cipher text, the receiver first uses its **private key (Kr_B)** to decrypt the **session key**.
2. The receiver uses the **session key** to get the **plaintext and the hash code**.
3. The receiver uses the same hash function on the plaintext to produce the hash value.
4. The receiver uses the **public key of the sender (Ku_A)** to decrypt the hash code to produce the hash value.
5. Finally, the receiver checks the equality of both the hash values to authenticate the message.

